

# Shaik Ibtihajulla Sha

Software Engineer | Full-Stack & AI-Native Systems | TypeScript, Python & LLM Agents

+46 763246025 | shaikibtihaj07@gmail.com | Linköping, Sweden

LinkedIn | GitHub | Portfolio

## PROFESSIONAL SUMMARY

Software Engineer with 2 years of hands-on experience across professional roles, internships, and engineering projects in backend development, data engineering, and applied AI. Currently building production backend systems at ScandVPN, and previously delivered ML-driven analytics at AxioGreen and data pipelines at Datafynder. Core strengths include **Python, TypeScript, Node.js, REST API design, SQL/PostgreSQL, ETL/ELT data pipelines, machine learning (forecasting and anomaly detection), LLM applications, RAG, and agentic workflows with tool calling and MCP**, backed by **Docker, Git, and CI/CD** for reliable delivery. Known for owning features end-to-end, from requirements and design to production deployment, and for treating data quality, validation, and reliability as core engineering concerns. Seeking a Software Engineer or Backend/AI Engineer role where I can build scalable systems and contribute from day one.

## EXPERIENCE

### Backend - Software Engineer

2026 – Present

ScandVPN - VPN Application, Sweden

Python | REST APIs | PostgreSQL | SQL

- Design, build, and maintain backend REST APIs powering a consumer VPN application, covering user accounts, authentication, and core app functionality.
- Developed subscription and billing-related services that manage plan status, user entitlements, and access to the VPN service.
- Built and integrated VPN server APIs used to manage server availability and connectivity, supporting a fast, one-click connection experience across a growing global server network.
- Designed and optimized relational database schemas and queries for users, subscriptions, and server data, prioritizing reliability, performance, and privacy-conscious data handling.

### AI Engineer Intern

Jan 2026 - Jul 2026

AxioGreen - Energy & Sustainability Tech

Python | Machine Learning | REST APIs | SQL

- Developed and integrated AI-driven features into an energy monitoring platform used to analyze electricity consumption across commercial buildings and healthcare facilities.
- Built machine-learning solutions for energy forecasting and anomaly detection using historical energy, sensor, and operational data.
- Developed Python-based APIs to expose model predictions to production applications, and integrated AI-generated insights into existing dashboards and energy-management workflows.
- Identified consumption patterns and abnormal usage, delivering actionable recommendations to improve energy efficiency.

### Data Engineer

Aug 2025 - Aug 2026

Datafynder - India

Python | SQL | ETL/ELT | BI & Dashboards

- Developed and maintained data pipelines processing VPN application, network, subscription, and user-interaction events.
- Built ETL/ELT workflows in Python and SQL that transformed raw product and operational data into analytics-ready datasets, and supported data warehouse and BI infrastructure.
- Created dashboards and analytical models to monitor VPN connection reliability, server performance, user engagement, subscription conversion, and retention.
- Implemented data validation and pipeline monitoring to improve data quality, and worked with engineering and product teams to turn operational data into actionable insights.

## PROJECTS

### AI Workflow Automation Platform

Python | LLM Orchestration | Tool Calling | OpenAI API | REST APIs

- Architected a workflow automation engine using LLM orchestration and tool/function calling to execute multi-step business workflows end-to-end.
- Integrated external REST APIs so agents can retrieve real-time data, trigger downstream actions, and return structured, rule-validated outputs.
- Designed a modular architecture with separate planning, execution, and validation stages for maintainability and production readiness.
- Added reliability guardrails (retries, fallback routing, structured error handling) and a lightweight evaluation framework to measure task accuracy.

**Production Enterprise AI Knowledge Platform (RAG)**

*Python / LangChain / OpenAI API / Vector Database / Embeddings / Docker*

- Built a Retrieval-Augmented Generation system that indexes internal enterprise documents into a vector store for precise, low-hallucination retrieval.
- Implemented embedding-based semantic search to return relevant document chunks and generate citation-grounded answers.
- Designed a document ingestion and chunking pipeline with metadata filtering, plus an evaluation layer for retrieval precision and answer faithfulness.
- Containerized the system with Docker and exposed it through a REST API with a modular, scalable backend architecture.

**Probabilistic User Behavior Modeling (Bachelor’s Thesis)**

*Python / Markov Models / Statistical Analysis / Synthetic Data*

- Designed first- and second-order Markov chain models to represent sequential user interaction patterns.
- Generated synthetic behavioral data to test the models and uncover latent structure relevant to product analytics and personalization.
- Documented methodology, experiments, and findings in a full academic thesis.

**Customer Churn Prediction System**

*Python / Scikit-Learn / Pandas / PostgreSQL / Feature Engineering*

- Built an end-to-end ML pipeline covering data cleaning, feature engineering, model training, and evaluation on PostgreSQL-backed data.
- Improved AUC-ROC over the baseline model through feature engineering and model tuning.
- Presented results in stakeholder-facing dashboards tied to actionable customer retention strategies.

**EDUCATION**

**Blekinge Institute of Technology (BTH), Sweden**

*2025 – 2026*

*Exchange Program in Computer Science*

**Relevant Coursework:** Machine Learning, Applied Statistics, Data Analytics, Statistical Modeling, Databases & SQL, Software Engineering, Python Programming

**TECHNICAL SKILLS**

**Languages:** Python, TypeScript, JavaScript, SQL

**Backend & APIs:** Node.js, FastAPI, Flask, REST APIs, Microservices, System Design

**Frontend:** React, Responsive UI, State Management

**Databases & Data:** PostgreSQL, SQL, ETL/ELT, Data Pipelines, Data Warehousing, BI Dashboards

**Applied AI & Agents:** LLMs, Tool/Function Calling, OpenAI API, Anthropic Claude API, OpenAI Agents SDK, MCP, RAG, Vector Databases, LangChain, LLM Evaluation, AI Guardrails

**Machine Learning:** Scikit-Learn, Pandas, NumPy, Feature Engineering, Forecasting, Anomaly Detection

**Cloud & DevOps:** Docker, Git, Linux, CI/CD, Microsoft Azure

**LANGUAGES (SPOKEN):**

English (Fluent), Swedish (Intermediate)